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REVERSE ENGINEERING AN APPLE WATCH INTO A WRIST WEAR MORE EFFECTIVE TO DEFY ALZHEIMER'S DISEASE

ABSTRACT

Despite various technology innovations, there are very few of which effectively assist individuals with Alzheimer's disease while being budget-friendly, usable, and convenient. Therefore, this project reworks the widely known Apple watch to a design that is very efficient for the concerned audience while embracing the addressed three characteristics. From a collaborative engineering design process grounded by research, including interviewing the target customers, the final product was established. In such a process, MATLAB is also used to analyze the product's cost and projected effect on helping individuals take their medications on time. By mathematically expressing its performance allows to examine for any relevant changes to be made to obtain the most efficient product. After 3 crucial iterations, the design is fixed to be a screen-free watch that has a panel primarily featured with voice commands, AI technology, and enhanced media response. These features make the device become more user-friendly to easily assist them navigate their days. Therefore, showing the design fulfilling customer satisfaction from meeting the important necessities of the concerned individuals, which are learnt through the interviews. Additionally, the product is ensured to be accessible for all by removing the insignificant luxuries from the Apple watch, which results in the cost of the product is projected to decline significantly. Moreover, the use of BOM was beneficial in representing the materialistics expenses and shaping a fairly accurate budget for manufacturing. Effective marketing strategies, like posters, are also created to further boost the accessibility of the product. Having said this, this project is evident to the power of reverse engineering as it shows that a new product is not always necessary to solve a critical problem in the world.

INTRODUCTION

About 55 million people in the world deal with memory loss and struggle with cognitive abilities throughout their daily lives [1]. This brain disease is known as Alzheimer's and is typically found in many elderly people. They face issues, like not being able to properly hold conversations, feeling challenged when using everyday items, and even forgetting to take their medicines on time. This is alarming. Over the years, many companies have innovated technologies, such as Lifetime Day Clocks, to make their lives easier [2]. However, such assistive devices are often found to be on the expensive side and lack efficiency. Many found it difficult to use and to rely on. Therefore, with the goal of resolving this, the project was initiated. It holds the purpose to reverse engineer a product that is effectively helpful for Alzheimer's patients and is accessible, sustainable, and user-friendly. That said, the Apple Watch is the ideal product to use as the foundation. Its exceptional and high-quality design proves to be essential in making people's lives more convenient and healthier.

BACKGROUND

Alzheimer's disease is a type of dementia that affects memory, thinking and behaviour. Symptoms eventually grow severe enough to interfere with daily tasks. Alzheimer's is not a normal part of aging. The greatest known risk factor is increasing age, and the majority of people with Alzheimer's are 65 and older. A person with Alzheimer's lives four to eight years after diagnosis but can live as long as 20 years, depending on other factors [4]. People with Alzheimer's often struggle to remember newly learned information, which is one of the earliest and most common symptoms. As the disease continues, they may experience confusion about time and place [4]. Some people living with changes in their memory due to Alzheimer's or other dementia may experience changes in their ability to develop and follow a plan or work with numbers[5]. As mentioned above, Alzheimer's is a disease that causes a slow decline in memory, thinking and reasoning skills[5]. Because of this, there should be a device that can help people remember daily tasks and routines.

A wrist watch, like an Apple Watch can be useful to give daily reminders and alerts to make life easier[6]. The watch will include only a few useful apps, so it's easy to use nothing complicated. It will have large icons, alerts for medication, and upcoming appointment reminders. This watch will not be just useful for the person with Alzheimer's but also for the people around them. I had a discussion with a family member who has difficulties remembering stuff. She mentioned that it's hard for her to remember what time she needs to take her medication or how many tablets she's supposed to take. She has a hard time using mobile phones or smart watches and said she would prefer a device that is easy to use and isn't too complicated.

The product that was chosen for reverse engineering is a smartwatch that is similar to the Apple Watch [6]. The watch can help make people's lives easier through constant reminders, alerts, and notifications. Furthermore, these types of watches help us understand how it works, how it helps, and what parts of the watch can be simplified to make it easy to use. The original Apple Watch is designed in such a way that it keeps time, sends notifications, tracks health data, and connects to your phone for calls, messages, and music. The watch also has a silent mode feature, which vibrates when there's a notification or any sort of alert [7].

By looking at this, the group was able to decide which features are helpful for the Alzheimer's patient and which one needs to be redesigned. Apple released its first-ever watch in 2015, which included fitness, notification, and quick communication on the wrist[8]. Over the years, Apple has made their watches better and advanced, such as longer battery life, stronger safety features and a password to unlock the watch.

Looking over the history, we see how the device has advanced and helps us make a more accessible design for the targeted audience. Apple Watch runs on a processor, touchscreen, sensors, and a battery that helps keep track of things. For our Alzheimer watch, we will produce a simplified version that includes a large icon and fewer but useful apps, making it user-friendly. Some materials in the BOM for the watch include: a battery, a processor, a speaker, wrist strap, and Bluetooth.

EVALUATION CRITERIA

Product Quality	*Grading (%)*	*Justification*
Efficiency The product can satisfyingly fulfill the primary needs of Alzheimer's patients while being productive and performing optimally.	99	The project aims to achieve 100% customer satisfaction because of its strong purpose and impact it holds on people's lives. It is projected to make the concerned audience's lives 3x more convenient and save $400 by removing the need for any external resources/assistance.
Durability/Sustainability The product can firmly withstand and operate adequately despite any unforeseen accidents, weather conditions, and any other rough impact.	95	The design passes various stress tests with a score above +90%. Tests impersonate common real-life scenarios to ensure it is resistant and strong. Also, it is designed to be 97% waterproof and scratchproof as well as to achieve a 4-point grade of flexibility. Therefore, it is known to be a long-lasting product, especially with a battery that lasts for 168 per full charge. Above all, 96% of materials used are eco-friendly and can be recycled in a healthy manner.
Cost-Effective The product can substantially function as intended and solve the addressed problem to a high degree at the most possible minimum cost.	99	Making the product be budget-friendly was a demanding priority for this project because of the goal to make it accessible to all. 100% of the modeled design is set to be manufactured without needing unreasonably

		expensive materials. In fact, the new design only requires ⅜ of the costs that is needed to make an Apple watch.
User-Friendly The product is straight-froward and can be easily used and understood by all people regardless of age, culture, knowledge, and more.	97	20+ hours of research was invested in thoroughly understanding the common issues people with Alzheimer's face regarding performing daily activities. The product is ensured to have about 0% limitations on the use, hence users can use the device to their liking either by voice or touch. It has a 1.8 seconds average response time with 0.2% lag when engaging with the design.
Safe for Users The product is recognized to not harm or put users' lives at risk at no costs and is labeled to be used without stress on safety measures.	100	The design passes with a 5 star grade on tests that analyzes for any risk in harm to users. In other words, 0 identified risk to human danger. 100% of the materials used are toxic-free and safe to interact with. Further, frequency levels can only be set between 60 db to 75 db to make sure audio levels are healthy for the human ear. The integrated AI technology is 98% accurate, thus ensuring close to absolutely no false information or assistance is provided.
Low-Maintenance The product requires few to none maintenance to function to its potential and be effective.	99	The design has a 1.05% chance to malfunction when in use by customers. Also, the integrated AI technology removes the need for individuals to manually update the device since it automatically does it for them. The AI can reliably self-fix about 99% of any internal problems.
High Comfort The product is comfortable to use and does not act as a hassle for users.	99	The concerned individuals already struggle a lot in their daily lives, therefore it is a priority for the product to not cause any discomfort. 100% materials used for the design are soft and it only weighs 45% of the weight of a standard Apple Watch. Essentially, giving them an almost weightless-like feeling on their wrist.

Equity and Fairness The product is a design for all and does not contain any bias or favour any specific ethnical/cultural, gender, or other types of people.	99	The product is 100% diverse and is not designed to suit only a certain group of people. It is designed with a general vision and does not consist of any special attributes that speak to only a few. In fact, the product has 0% limitation on the setting/location it can be used. Hence, making it fair for all who live in different environments.
Impactful Marketing Eye-catching marketing strategies are used to publicize the product and bring awareness.	99	Engaging and trending methods of advertisements are used to inform the public for at least 95% of the customers to consider this product as an option. This is key in making the product more accessible and ensuring the product speaks to the concerned individuals. .

REDESIGN PROPOSAL

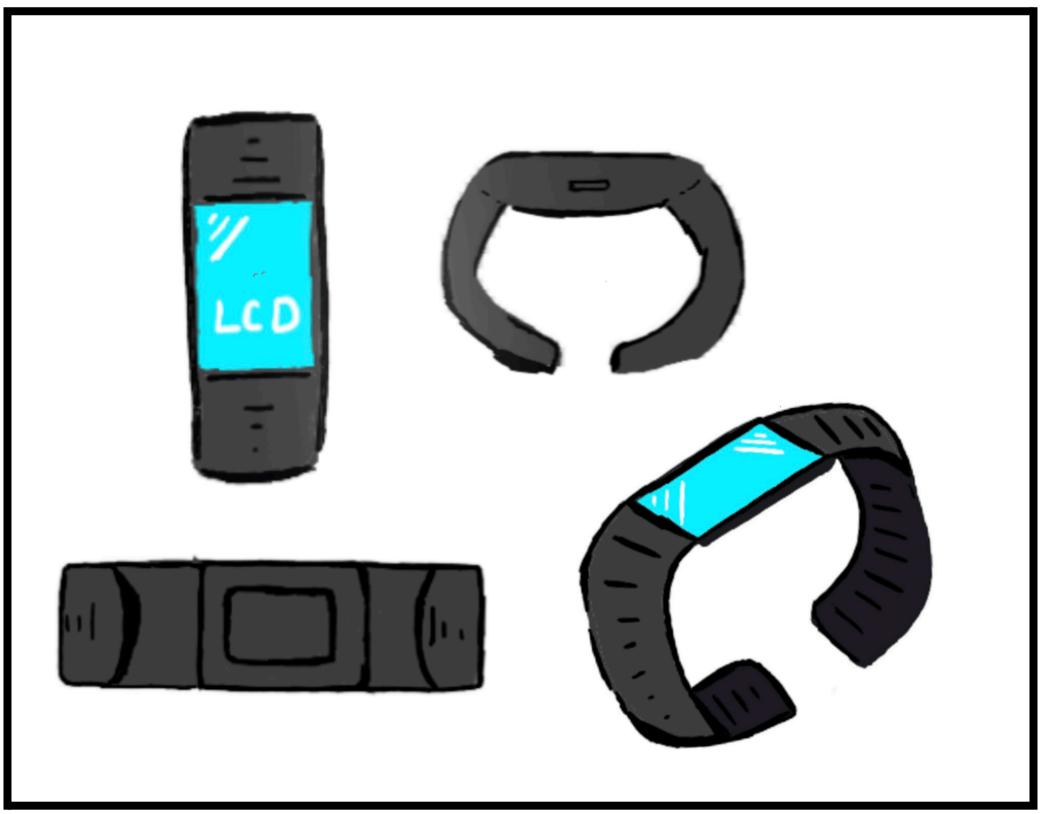

Figure 1: *First design*

The different angled sketches above express the first iteration of the redesigned Apple watch and provide a visual representation of the direction the design is planned to head towards.

The first prototype of our reverse engineering smartwatch is our first initial attempt to modify the design of the Apple Watch, aiming to help address the needs of people with Alzheimer's, while cutting down on unnecessary features while also maintaining affordability. This first reiteration closely resembles a modern smartwatch which includes an LCD screen, a silicon strap for comfort and hygiene, and charges via USB-C for universal compatibility. These features are basic frameworks for most modern smartwatches but come with a price of being very complex to use because it relies on visual touch inputs and being costly.

We introduced an AI-infused Siri module and a flashing alert light, which help simplify the user experience through voice control capabilities, as well as multi-sensory notification methods. These features help with struggles in memory loss, communication, and to grab the user's attention visually. Although these features were a good first step. The inclusion of the LCD screen made the device too intricate and overwhelming to use, and too expensive for the specific needs of people with Alzheimer's. The first design highlighted the need to simplify the screen further, while meeting our goals of making it budget friendly and easy to use.

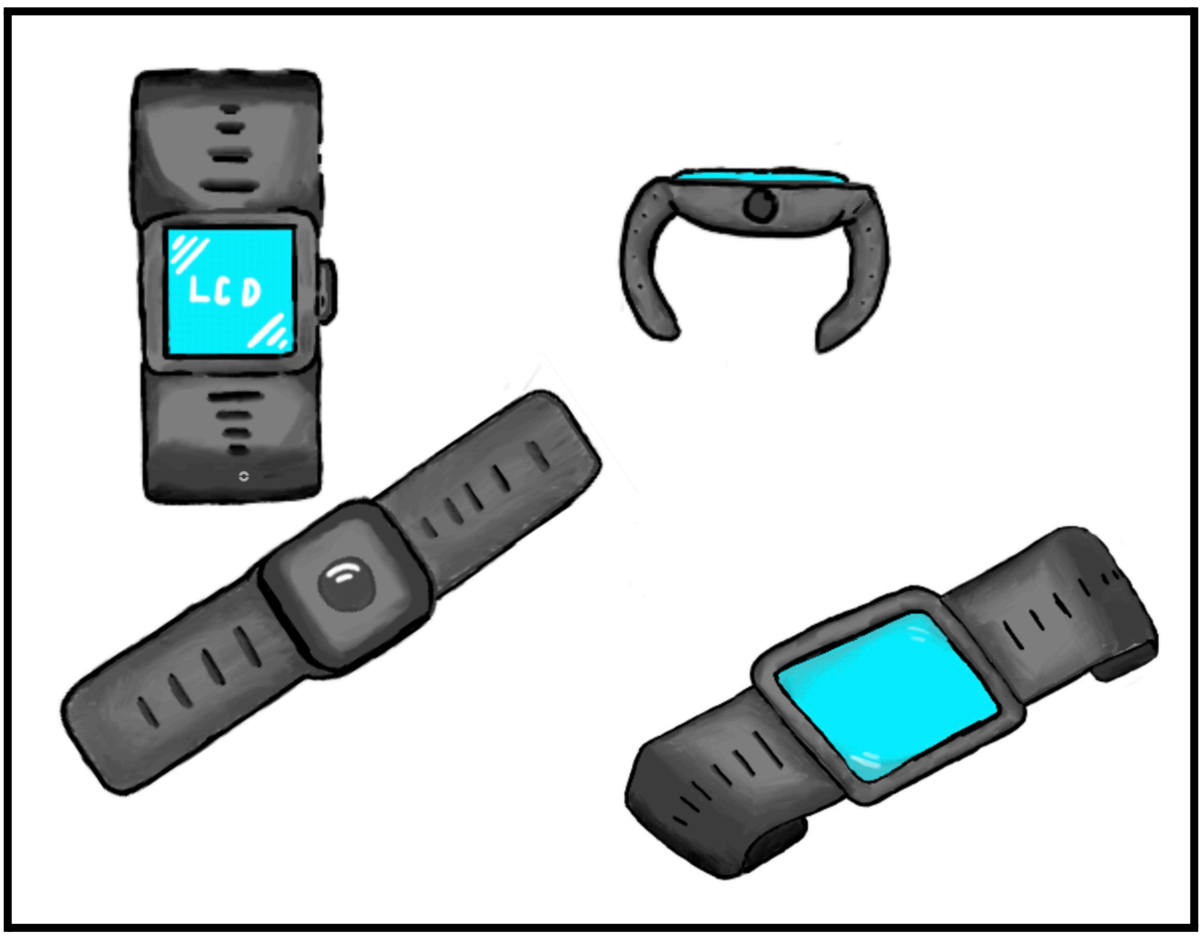

Figure 2: *Second design*

The sketches above similarly displays the sketches of the watch with respect to different views, but for such iteration more necessary features are added, while more unnecessary ones are continued to be removed.

The second prototype of our smartwatch implemented physical controls and refined ergonomics, making it more user friendly. We changed the physical design of the watch by adding wireless charging, screen reduction, and tactile control. The change from manual charging to wireless charging was made to help with limited dexterity and memory, which removes the frustration of losing charging cables and plugging in cables. Reducing the screen by making it smaller with smaller bezels helps with manufacturing costs and makes the device less visually overwhelming to use than the first prototype. The addition of a side button was made to give the user a physical and tangible way to navigate the menus of the smartwatch.

Furthermore, the second design focuses upon making sensory and communication improvements. Voice clarity is improved by making the microphone noise cancelling. This is needed because the watch needs to filter out background noise fully, so the voice AI commands work to its best potential. In addition, a larger speaker is implemented to help address hearing issues in people with Alzheimer's, ensuring that alerts and voice commands are heard. Although the audio levels are increased, it was ensured they are still kept at a frequency that is safe for hearing during a long period of time. Therefore, a maximum of 75 db is implemented with users being able to change the audio level to their liking. In addition, a larger haptic feedback motor is included, as standard vibrations or touch can be missed by most people. The second design solidified the redesign by mainly focussing on achieving sensory components as well as shifting our focus on improving upon the visual aspects.

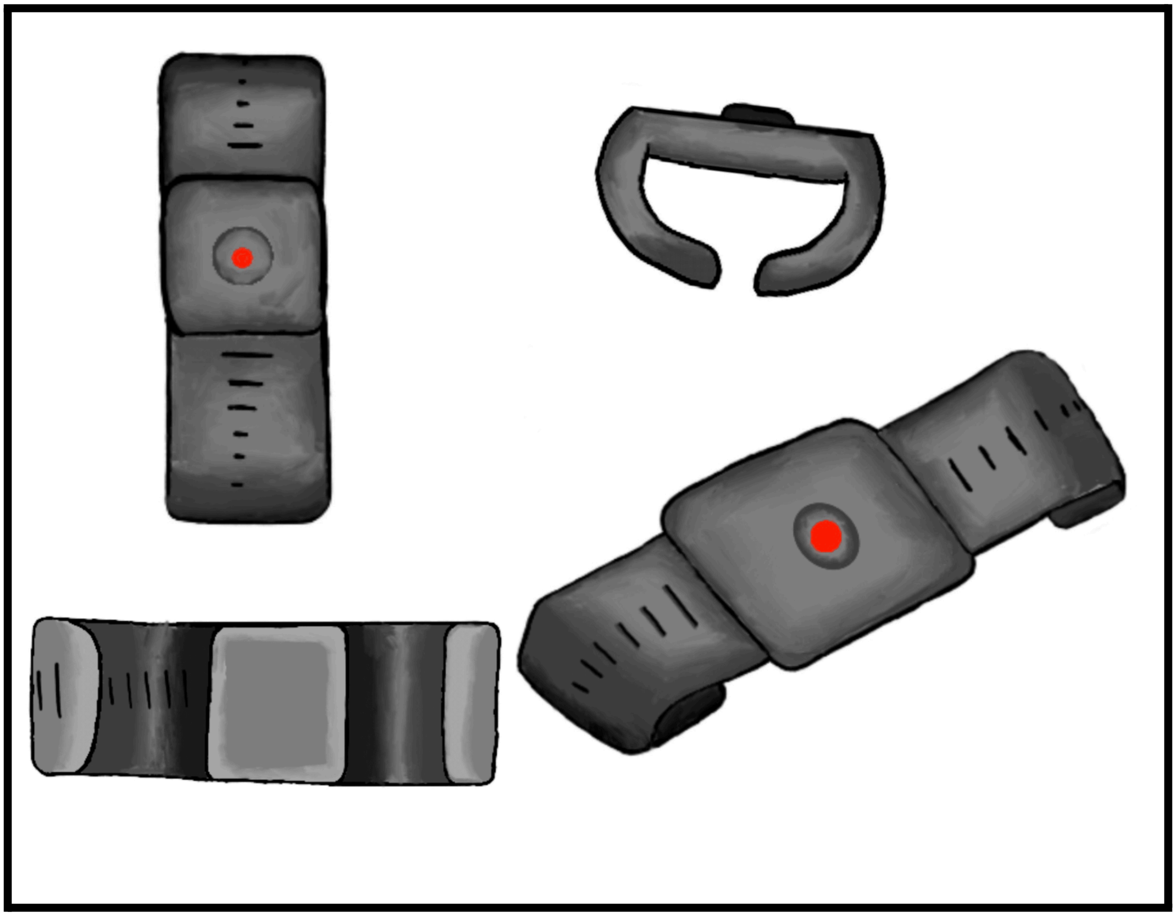

Figure 3: *Final design*

The sketches show the finalized design in a manner similar to the other figures and are sketched in detail to elevate accuracy of the team's vision.

The final prototype of our smartwatch includes simplifying the visual components of the device even further by getting rid of the LCD screen entirely. This eliminates the primary visual confusion for people with Alzheimer's, while also reducing manufacturing costs significantly. Additionally, it added simplicity to the device through the removal of the side button. As the red blinking light now acts as both a button and a visual indicator. It is important for the product to be as simple as it could be as the team intended it to be something that can be used by all regardless of knowledge or skill level. In fact, it also makes it more user-friendly because users can simply just wear the watch to use it instead of manually navigating a small screen to set certain settings.

This design also includes replacing the wireless charging entirely and replacing it with a larger battery. While wireless charging is convenient, a battery that can last for approximately a week is far more valuable than a device that needs to be charged daily for individuals with Alzheimer's. This lowers the maintenance required for the product, which overall boosts customer satisfaction. Removing the screen and wireless charging gave the required space needed to place a larger battery in the device. The simplicity of the device is found in the red blinking light, as it becomes the only visual cue for urgent notifications, serving as a clear indicator for the user or caregiver. In other words, the blinking light is far more effective in warning individuals when compared to a digital notification tab that is likely to be clustered on the screen.

MARKETING

The watch is a user-friendly smartwatch that is specifically designed for elderly individuals diagnosed with Alzheimer's or dementia. It allows aging adults to utilize simple reminders or alerts, and AI task management, promoting independence while protecting individuals. By eliminating unnecessary luxury features from standard smartwatches, the watch focuses on affordability, accessibility, and usability.

Target Market

The primary target market consists of seniors with Alzheimer’s or dementia who require assistance with daily tasks such as taking medication, remembering appointments, or managing schedules. Whereas, the secondary target market includes family members and caregivers supporting seniors, as well as health care providers, retirement homes, and senior care facilities looking for cost-effective, user-friendly technology solutions.

Marketing Strategy

The marketing strategy emphasizes targeted communication and partnership development with the goal of building awareness of the watch benefits among seniors and their caregivers. It highlights usability, independence, and affordability as key selling points while leveraging partnerships with healthcare organizations, retirement homes, and non-profits to reach the intended audience effectively.

The approach includes social media campaigns targeting adults (ages 30 - 60+) and caregivers on platforms such as Facebook and Instagram. In addition, community and healthcare outreach involve product demonstrations at retirement homes, and pharmacies. Local media, such as print advertisements like newspapers, and radio segments help spread the message while focusing on trust, simplicity, and product benefits. Finally, building partnerships with government programs and non-profit organizations aim to increase access for low-income seniors by sponsoring programs

Marketing Approach and Message

The core message is: “Independence, reminders, and peace of mind, all on your wrist.”

The advertising approach emphasizes that by cutting the number of features on the device, it becomes more affordable, while also ensuring full accessibility of the drive. This aspect of the device is based on the actual “lived experience” of real users. For example, when an interview was conducted with Margaret, an older adult, she shared her hesitancy about navigating potentially complex menus in a smart watch. Most importantly, she said “I just wish I could do everything by voice with the watch.”. Users also complained that charging the watch each day was difficult to keep up with. Her comments emphasize the idea of simplified access. In response to her comments, the idea was proposed that the watch would include AI-enhanced accessibility control, louder alerts, longer battery and prompts to increase all of the access features for reminding of tasks and events.

On the advertising side, showing similar relatable stories of seniors is very important, so creating newspapers of seniors using the watch confidently in their day-to-day lives is essential. The messaging highlights the practical benefits it provides; flashing reminders, haptic alert functionality, faster accessibility with AI-scheduling, and longer battery life to demonstrate that concerns were listened to and addressed the challenges as expressed by participants like Margaret, through design improvements. Moreover, testimonials from other seniors and from caregivers were included to enhance the device's trust and credibility. Woven into the advertisement is the notion of conveying the practicality of the device in emotional ways.

The marketing strategy embraces clear messaging, empathy, and accessibility. The approach is diverse, reaching both seniors and caregivers with a focus on independence, simplicity, and reliability. The partnership gives clinical healthcare providers additional pathways to distribute the device technology to anyone who may benefit from it. Affordable, easy-to-use, high-functionality matters. The watch focuses on easy to use with built-in support for aging adults and their families.

Bill of Materials (BOM)

The Bill of Materials (BOM) outlines all components necessary to assemble the device, along with realistic estimated costs per unit. Each component is selected to optimize functionality, usability, and reliability of the device, as well as to account for a prototype production

Table 1. *Materials, cost, and Quantity (Upgraded Features)*

Item No.	Component	Description & Purpose	Quantity per Unit	Est. Unit Cost ($ CAD)	Notes
1	Silicone Strap	Soft, hypoallergenic band	1	$5.00	For comfortability and affordability
2	Larger Battery	Long-life rechargeable battery (week-long)	1	$15.00	Determine capacity needed
3	Better Microphone	High sensitivity, noise cancelling microphone	1	$8.00	Check pickup patterns
4	Louder Speaker	High-volume speaker for audio output	1	$10.00	Check frequency response
5	Larger Haptic Feedback Motor	Strong vibration motor alerts	1	$6.00	Ensure compatible voltage/current

6	Flashing Alert Light	Visual alert light (LED)	1	$2.00	Choose low-power LED
7	AI-Infused Siri Module	Software + hardware module for task/reminders	1	$20.00	Firmware included (embedded in phone)
8	Application Software	App to create/modify/delete reminders	-	$5.00	Development over units, low per-unit cost
9	Packaging & User Manual	Eco-friendly box with large print instructions	-	$3.00	Cost effective, visually accessible packaging
10	Assembly & Quality Testing	Final product testing and verification	-	$10.00	Maintaining a reliable device
11	Marketing & Distribution Cost	Ads, logistics, outreach (ex. Newspapers, social media, etc…)	-	$15.00	Includes social media campaigns, brochures, shipping
-	**Total Unit Cost**	-	-	*$99.00*	-

Total Production Cost per Unit: $99.00
The sum of all components, packaging, assembly, software and marketing per unit

Projected Retail Price: $125.00
for affordability and sustainability compared to Apple Watch [3]
Calculated by Unit Cost x (1 + Profit Margin)

Profit Margin: 25%

The components selected for the BOM were carefully evaluated for performance of the device, practical construction, and prototype-scale constraints. The haptic feedback motor, vibrations and flashing light provide multi-sensory notifications to the user, allowing them to recognize notifications in different conditions and environments. The microphone and speaker allow for improved audio input and audio output when dictating and receiving notifications. The AI-Infused Siri module, along with application software, combines hardware into a module for tasks and managing reminders..

Additional costs for manufacturing, assembly, and packaging hypothetically represents the realistic requirements for prototype-level production. The total estimated unit cost of $99.00 provides a realistic basis of what the unit can cost for budgeting, which considers the overall quality of components while utilizing low-volume assembly and basic hardware integration in a prototype scenario. While this unit estimate is realistic in the development stages, it will vary when there is pricing, availability, and volume of the final product.

MATLAB ANALYSIS

Analysis of Watch Alert Effectiveness (Data analysis and visualization)

Overview

The purpose of this program is to determine the effectiveness of our watch's medication reminding ability. To determine this, we analyzed three different scenarios: no-watch, Apple Watch, and our watch. We hoped that our stronger vibrations and LED lights would be significantly better than the Apple Watch. We included the no-watch scenario to determine if the watches made a difference in medication taken. To keep our data consistent, we limited it to the following parameters:

- The patient will take three doses of medication.
- We will only record one week of data.

Using this data, we analyzed the total amount of medication taken on time and tallied the results. The results would be later used to generate the bar graph below.

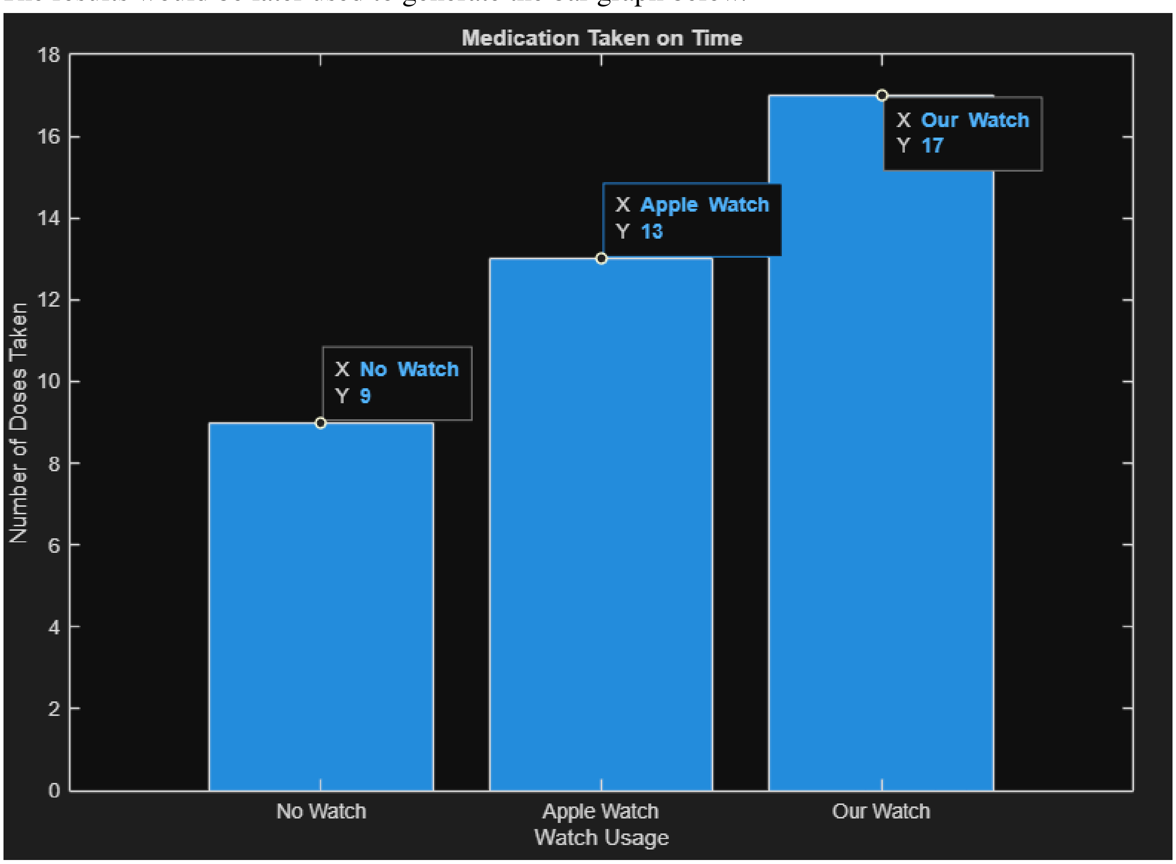

Figure 4: *Medication on Time*

Figure 4 visualizes the effectiveness of various watch reminders

X-axis: Watch Usage
Y-axis: Number of doses

Observations:
Highest performance: Our watch
Lowest performance: No watch

<u>Conclusion</u>

Based on the results, we can observe that both the Apple Watch and our watch greatly exceed the results of no watch. With this information, we can conclude that this is a viable solution to reminding Alzheimer's patients, justifying our product. Comparing the Apple Watch with our watch, we can see a clear difference in performance, with our watch reminding the user five more times a week than the Apple Watch. This supports our design decision to implement stronger vibrations and LED lights compared to the Apple Watch.

Bill of Materials (BOM) and Cost Analysis

<u>Overview</u>

The Bill of Materials contains an entire list of parts needed for the watch and includes the approximate cost per unit. These parts are not only reliable, but they are also very affordable. The BOM hardware includes (straps, screen, battery, haptic motor, etc.). Moreover, the software includes (AI Siri module and a reminder app. Other costs include assembly/testing and packaging. On a side note, marketing costs were not included in the component because it does not add value to the physical product.

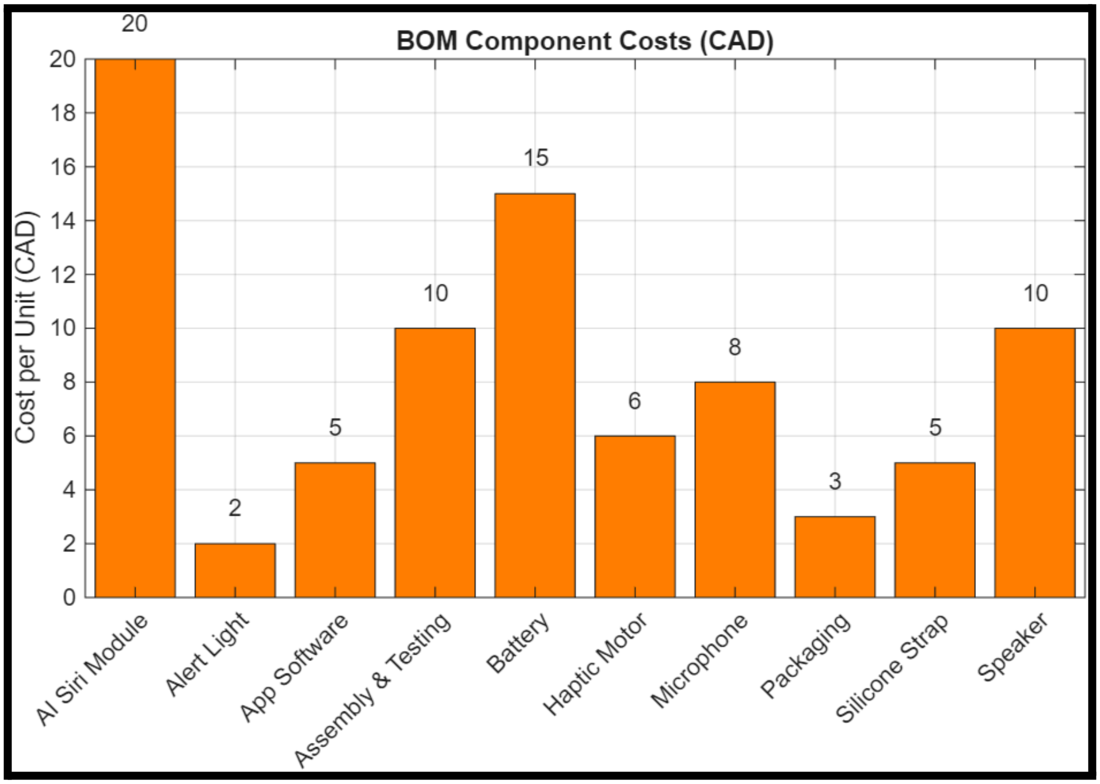

Figure 5: *Component Costs Bar Graph*

The bar graph in figure 5 visualizes the cost of each component:

X-axis: Components of the watch
Y-axis: Costs per Unit

Observations:
The most expensive component: AI Siri ($20), and Battery ($15)
The most inexpensive components: Alert light ($2), Side Button ($3) and Packaging ($3)

When the BOM is added up, we get a total unit cost of $112 CAD, which is equivalent to the total value of the components.

<u>Conclusion</u>

This chart can be used to justify cost allocation and in making design decisions that could decrease the total cost per unit cost where applicable.

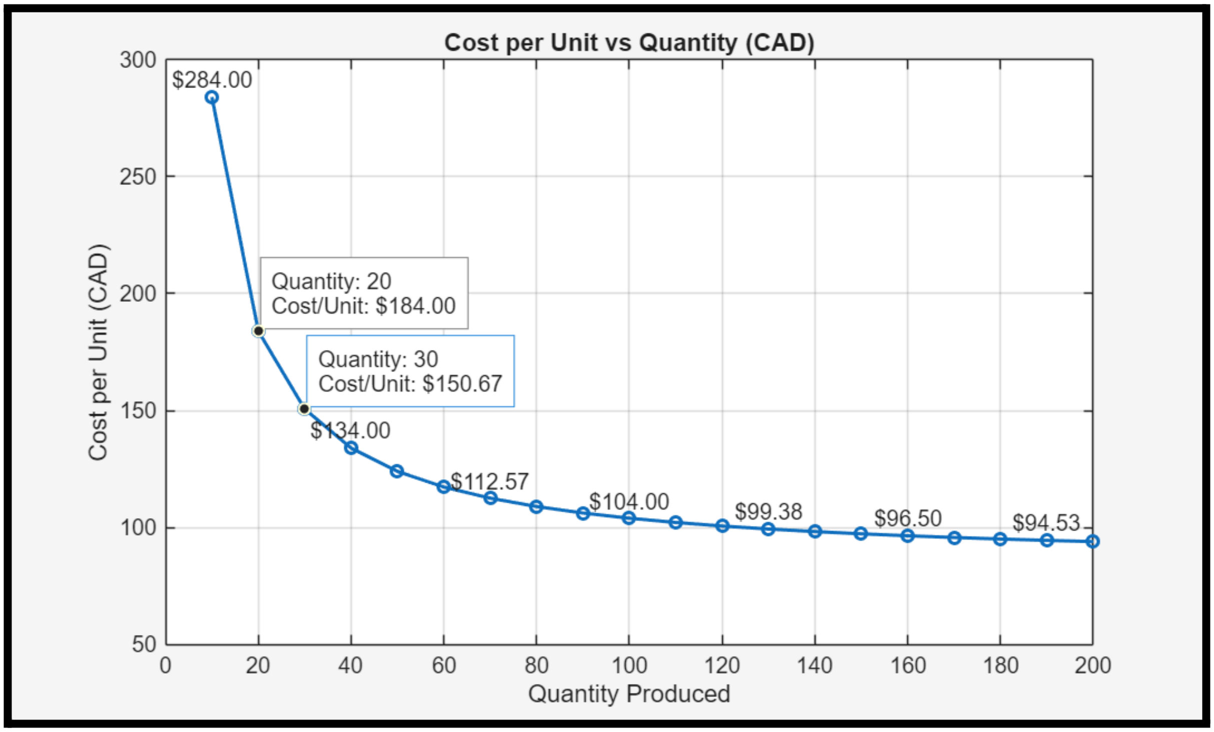

Figure 6: *Cost vs production quantities*

Figure 6 illustrates the relationship between production quantity and cost per unit:

X-axis: Number of units produced
Y-axis: Cost per unit

Labels: Every third point is labelled for readability. Furthermore, the chart includes interactive data that allows exact (x,y) values to be viewed dynamically.

Observations:
When the production quantity of the watch is low, the cost to produce one unit is higher as the tooling cost ($2000) is spread amongst fewer watches. However, when production quantity is greatly increased, the tooling cost is distributed over a greater number of units, reducing the fixed cost per unit and lowering the average build price of one unit overall.

Unit cost calculation:
The unit cost of production (average cost per unit) is found through the following formula:
Unit cost of production = (Tooling Cost + Unit Cost x Quantity)/Quantity. This formula takes into account the fixed costs (tooling) and variable costs (components and assembly). It visualizes the decrease in cost of the watch, highlighting how mass production benefits users and profit margins.

Design Decisions and Cost Optimization

This watch is simplified compared to an expensive Apple Watch[3]:

- No screen (simple touch features)
- Smaller haptic motor
- Cheaper silicon strap rather than a high-quality band
- Less unnecessary software features

Having lowered the unit cost, the initial starting budget can be used to produce more units. This makes the watch more accessible in early stages, increasing awareness and driving sales. Increased awareness makes the watch more accessible, fulfilling the purpose of the product.

Total Summary of Matlab Analysis

The Watch alert effectiveness bar chart highlights the change in medication use following the use of the Apple Watch and the proposed smart watch. The BOM Chart indicates the distribution of costs and highlights the most costly parts that have been streamlined to create this watch. The Cost vs Quantity plot highlights the importance of production quantity. Streamlining or reducing parts reduces the unit cost, which allows for more production with the same budget, lowering our average cost, thus raising more awareness for the watch. This lower production cost, in combination with the analysis of the watch's effectiveness, highlights the simpler, more accessible watch created for users with Alzheimer's, which considers functionality, usability, and affordability.

CONCLUSION

In conclusion, this project successfully demonstrates how reverse engineering can transform an existing device into a highly accessible and meaningful solution for individuals living with Alzheimer's disease.

By redesigning the Apple Watch into a simplified, screen-free wrist device, the team served real issues expressed by interviewed customers of the Apple Watch [3]. The final design on the real needs expressed by users during interviews-such as difficulty navigating complex menus, remembering medication times, or maintaining daily routines. The final design achieves this by removing software and hardware features that fail to serve Alzheimer's patients, and prioritizes the features that help the elderly. This is evident in the bar chart, which highlights the uptrend in medication usage when using the watch.

Furthermore, the project prioritizes accessibility and affordability by removing unnecessary features and utilizing cost-effective materials. This is evident within the cost analysis, as features are highlighted and unnecessary luxuries are removed from the Apple Watch, making the watch budget-friendly [3]. This is done by reducing the cost of production, allowing us to increase the initial number of produced watches and raise awareness for the proposed product.
Overall, this project highlights the impact of reverse engineering: meaningful innovation does not always require creating a completely new device, but rather understanding users deeply and enhancing existing technology to better serve them. The final design is a practical, safe, and user-centred solution that has the potential to improve the quality of life for individuals with Alzheimer's and provide peace of mind for their families and caregivers.

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